



Smart Traffic RL

User Manual & Setup Guide

Multi-Agent Reinforcement Learning for Intelligent Traffic Control

Project: Smart Traffic Signal Control with MARL

Environment: Python 3.11 + .venv Virtual Environment

Simulator: SUMO 1.26.0

RL Framework: Stable-Baselines3 + PettingZoo

Dashboard: Streamlit + Plotly

Table of Contents

1. Prerequisites & System Requirements
2. Environment Setup (Which Environment to Choose)
3. Activating the Virtual Environment
4. Running the Project
5. Menu Options Explained
6. Recommended First-Time Run Order
7. Direct Command-Line Usage
8. Troubleshooting

1. Prerequisites & System Requirements

Before running the project, ensure the following software is installed on your system:

Requirement	Required Version	Purpose
Python	3.10 or 3.11 (NOT 3.14)	Core runtime
SUMO Simulator	1.19+ (recommended 1.26)	Traffic simulation engine
pip	Latest	Package manager
PowerShell	5.1+	Terminal (Windows)

CAUTION

Your system has Python 3.14 as default, but PyTorch does NOT support Python 3.14. The project includes a .venv virtual environment with Python 3.11.8 which you MUST use.

Installed Packages (in .venv)

All required packages are pre-installed in the .venv virtual environment:

Package	Version	Purpose
torch	2.11.0	Deep learning framework
stable-baselines3	2.8.0	RL algorithms (PPO)
gymnasium	1.2.3	RL environment interface
pettingzoo	1.24.1	Multi-agent environments
supersuit	3.9.0	Multi-agent wrappers
streamlit	1.56.0	Web dashboard
plotly	6.7.0	Interactive charts
numpy	2.4.4	Numerical computing
pandas	3.0.2	Data processing
traci	1.26.0	SUMO Python interface
matplotlib	3.10.8	Static plotting

2. Environment Setup

Which Environment to Choose?

Your project directory contains multiple virtual environment folders. Here is what each one is and which one you should use:

Folder	Python Version	Status	Use This?
.venv	3.11.8	All packages installed	YES
venv	3.11.8	Incomplete setup	NO
venv_old	Unknown	Outdated / broken	NO
System Python	3.14.0	Incompatible with PyTorch	NO

CAUTION

ALWAYS use the .venv folder (Python 3.11.8). This is the only environment with all dependencies correctly installed. The system Python 3.14 will cause import errors with PyTorch and other packages.

Why Python 3.11?

PyTorch (the deep learning backend) and several scientific computing libraries require Python 3.10-3.12. Python 3.14 is too new and not yet supported by these libraries. Python 3.11 offers the best compatibility and performance.

SUMO Simulator Path

SUMO is installed at the following location. The main.py script automatically configures this:

```
SUMO_HOME = C:\Program Files (x86)\Eclipse\Sumo
SUMO_BIN  = C:\Program Files (x86)\Eclipse\Sumo\bin
```

3. Activating the Virtual Environment

Follow these steps every time you want to run the project. You must activate the virtual environment before running any Python command.

1 Open PowerShell / VS Code Terminal

Open a terminal window. In VS Code, press Ctrl+` (backtick) to open the integrated terminal.

2 Navigate to the Project Directory

PowerShell

```
cd "c:\Users\vipin\Desktop\NITIN_SARAJ\projects\ml_project\smart-traffic-rl"
```

3 Activate the .venv Environment

PowerShell

```
.\.venv\Scripts\Activate.ps1
```

After activation, your terminal prompt will change to show (.venv) at the beginning:

```
(.venv) PS C:\Users\vipin\Desktop\...\smart-traffic-rl>
```

4 Verify the Python Version

Verification

```
python --version  
# Expected output: Python 3.11.8
```

TIP

If you get an 'execution policy' error when activating, run this command once:
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

4. Running the Project

Option A: Interactive Menu (Recommended)

The easiest way to run the project. This shows a menu with all available options:

Command

```
python main.py
```

The menu will display:

Interactive Menu

```
+-----+
|                MAIN MENU                |
+-----+
|  1. Train Single Agent (PPO with Curriculum Learning)  |
|  2. Train Multi-Agent (MAPPO for 9 Intersections)      |
|  3. Run Dashboard (Real-time Visualization)           |
|  4. Run Evaluation Only                               |
|  5. Run Adversarial Tests                             |
|  6. Generate SUMO Network                             |
|  7. Run All Tests                                     |
|  8. Exit                                               |
+-----+
```

Type a number (1-8) and press Enter to select an option.

Option B: Command-Line Flags (Quick Access)

You can bypass the menu by passing a `--mode` flag:

Direct Commands

```
python main.py --mode train      # Train single agent
python main.py --mode multi     # Train multi-agent
python main.py --mode dashboard  # Launch dashboard
python main.py --mode eval      # Run evaluation
python main.py --mode test      # Run all tests
```

5. Menu Options Explained

Option 1: Train Single Agent

Trains a PPO (Proximal Policy Optimization) agent on the center intersection (n11) using curriculum learning. The training progresses through 3 stages of increasing difficulty:

Stage	Timesteps	Max Vehicles	Difficulty
Stage 1	50,000	100	Easy
Stage 2	100,000	300	Medium
Stage 3	150,000	500	Hard

Estimated time: 30-60 minutes. Model saved to: `agents/ppo_final.zip`

Option 2: Train Multi-Agent (MAPPO)

Trains 9 independent MAPPO agents (one per intersection) on the full 3x3 grid. Uses the single-agent model as a starting point if available (transfer learning). Each agent manages its own traffic signal while cooperating with neighbors.

Estimated time: 1-3 hours. Model saved to: `agents/mappo_final.zip`

WARNING

Train the single agent FIRST (Option 1), then the multi-agent (Option 2). The multi-agent system uses transfer learning from the single agent's weights.

Option 3: Run Dashboard

Launches a Streamlit web dashboard for real-time monitoring. The dashboard shows traffic metrics, signal phases, queue lengths, CO2 emissions, and explainable AI decision rationale. Opens at `http://localhost:8501` in your browser.

Press Ctrl+C in the terminal to stop the dashboard.

Option 4: Run Evaluation

Evaluates the trained models against a fixed-timing baseline to measure improvement.

Option 5: Run Adversarial Tests

Tests system robustness against: lane blockages (accidents), sensor failures, and multiple emergency vehicles simultaneously.

Option 6: Generate SUMO Network

Creates the 3x3 intersection grid network files. Run this FIRST if the simulation files (`grid_network.net.xml`, `routes.rou.xml`) don't exist yet.

Option 7: Run All Tests

Executes the complete test suite: SUMO connectivity, environment validation, and adversarial robustness tests.

6. Recommended First-Time Run Order

When running the project for the first time, follow this exact order. Each step depends on the previous one:

1 Activate .venv

Sets up the correct Python 3.11 environment

2 Generate SUMO Network

Creates the 3x3 grid simulation files

```
python main.py · Option 6
```

3 Train Single Agent

Trains PPO on center intersection (~30-60 min)

```
python main.py · Option 1
```

4 Train Multi-Agent

Trains 9 MAPPO agents on full grid (~1-3 hrs)

```
python main.py · Option 2
```

5 Launch Dashboard

View results at <http://localhost:8501>

```
python main.py · Option 3
```

6 Run Tests

Verify adversarial robustness

```
python main.py · Option 5 or 7
```

NOTE

Steps 3 and 4 (training) are computationally intensive. A GPU will significantly speed up training. If you only have a CPU, expect longer training times but the system will still work correctly.

7. Direct Command-Line Usage

Quick Reference Commands

Copy-paste these commands after activating `.venv`:

All Available Commands

```
# Interactive menu
python main.py

# Train single agent
python main.py --mode train

# Train multi-agent
python main.py --mode multi

# Launch dashboard
python main.py --mode dashboard

# Alternative dashboard launch
python run_dashboard.py
python -m streamlit run dashboard/app.py

# Run tests
python main.py --mode test
```

Project File Structure

Project Structure

smart-traffic-rl/	
main.py	Main entry point (run this)
config.py	Central configuration
requirements.txt	Package dependencies
run_dashboard.py	Dashboard launcher shortcut
.venv/	Virtual environment (Python 3.11)
agents/	Trained model files (.zip)
env/	Gymnasium & PettingZoo environments
training/	Training scripts & results
simulation/	SUMO network & route files
dashboard/	Streamlit visualization app
tests/	Test suites

8. Troubleshooting

Common issues and their solutions:

Problem	Cause	Solution
ModuleNotFoundError	Forgot to activate .venv	<code>.\. venv\Scripts\Activate.ps1</code>
Python 3.14 showing	Wrong Python active	Activate .venv first
Execution policy error	PowerShell restriction	<code>Set-ExecutionPolicy -Scope CurrentUser RemoteSigned</code>
SUMO not found	Missing simulator	Install from sumo.dlr.de
Dashboard won't open	Port conflict	Open <code>http://localhost:8501</code> manually
Training is very slow	No GPU / normal behavior	Use GPU if available; CPU works too
Permission denied	Admin rights needed	Run terminal as Administrator
traci error	SUMO_HOME not set	<code>main.py</code> sets this automatically

TIP

The most common issue is forgetting to activate .venv before running commands. Always check that (.venv) appears in your terminal prompt before running python.

Getting Help

If you encounter issues not covered here, check the README.md in the project root for additional documentation, or review the error messages carefully - they often indicate exactly what is missing or misconfigured.